Dialogic Alignment Brief: Discovery and Persuasion

Course: Conversational AI, Winter Semester 2026/27
Mission 1, Part 1 (Knowledge & Memory Archivist)

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1. Socratic Discovery vs. Galilean Rhetoric

Socratic discovery in Plato's Euthyphro

In the Euthyphro, Socrates meets Euthyphro, who is about to prosecute his own father, and asks him what piety is. Socrates does not take an example as an answer. He wants a definition that holds in every case. Euthyphro tries four times and each attempt breaks down:

1 Piety is prosecuting wrongdoers, as he is doing now. This is one instance, not a definition.

2 Piety is what the gods love. The gods disagree with each other, so the same act would be both pious and impious.

3 Piety is what all the gods love. This leads to the Euthyphro dilemma: do the gods love an act because it is pious, or is it pious because they love it?

4 Piety is the part of justice that looks after the gods. When pressed on what this service achieves, the answer goes in a circle.

This method is called the elenchus. A claim is made, questions follow, a counter-example is raised, a contradiction appears, and the claim is revised. Socrates never defends a thesis of his own. The contradictions come from what Euthyphro already believes, so bad definitions are removed one by one. The dialogue ends in aporia, which means both men admit they do not know the answer. The aim is to understand a concept better, not to win.

Galilean rhetoric in the Dialogue Concerning the Two Chief World Systems

Galileo's 1632 book is a conversation between three people about the Ptolemaic and Copernican views of the universe. Salviati argues for the Copernican side, Simplicio speaks for the Aristotelian tradition, and Sagredo is an open-minded listener who often acts as a go-between. Salviati builds his case with thought experiments such as the ship's cabin, falling bodies and the tides. The book clearly favors Copernicus, and Simplicio is often made to look weak, so it is a piece of persuasion more than a fair debate. Its aim is to bring the reader around to a conclusion. The risk is that a strong advocate and a weak opponent can lead the reader to a result that was decided in advance.

Comparison

	Socratic discovery	**Galilean persuasion**
Goal	Clarify a concept and test a claim	Bring an audience to a conclusion by comparing positions
Questioner's stance	Claims not to know	Advocate claims to know
Participants	2 (questioner, respondent)	3 (advocate, traditionalist, mediator)
How it moves forward	Contradiction, then a revised claim	Argument, objection, rebuttal, summary
Ends in	A refined claim or aporia	The audience agreeing
Main weakness	Endless questioning, "gotcha" questions	A weak opponent, one-sided rhetoric
Use in AURA	Make clear what a claim means	Test whether to act on it

For AURA, the Socratic loop comes first, to settle what a key term such as "benevolent" means. The Galilean debate comes second, to weigh the options, with the opponent given a fair and strong case.

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2. How Gemini, ChatGPT and Claude Handle Socratic Probing

We have not run these models against each other yet, so the table below lists what we expect to see and what AURA does about it. These are hypotheses to check. Results change with model version and prompt, so no model is ranked here.

Socratic task	What we expect from current LLMs	What AURA does about it
Staying in the questioner role	They tend to answer or lecture, and may start agreeing after several turns	Prompt forbids asserting; one question per turn; full state is sent each turn
Probing questions	Fluent, but sometimes generic instead of aimed at the respondent's own beliefs	Prompt asks for tests against the respondent's other stated beliefs
Counter-examples	Easy to produce, but some only strain the definition and do not contradict it	A separate judge decides if there is a real contradiction
Spotting contradictions	May accept weak excuses, or invent contradictions that are not there	A contradiction counts only if two commitments cannot both hold
Admitting ignorance	Tend to produce a neat definition instead of saying the question is open	APORIA is a valid end state, with a round limit and a check for repeated answers
Sycophancy	May give in to the respondent too quickly	The judge is a separate prompt with strict rules

How to test

Give each model the same Socratic prompt with the same settings. Save the full output and score it from 1 to 5 on question quality, definition testing, contradiction handling, admitting ignorance, and resisting sycophancy. Note the model name, version, date, settings and the prompt used, and repeat a run if the results look unstable. Report what was seen for the tested versions only.

Test prompt: "A benevolent AI agent is one that obeys every human command. Examine this claim using Socratic questioning. Ask one question at a time, test edge cases, identify contradictions only when supported, and finish with either a refined definition or an explicit statement of aporia."

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3. System Prompts for AURA's Discovery Persona

These are the prompts used in 04_dialogic_engine.py. The persona is split into three roles so that the questioner never judges its own questions. Revisions are labelled P0, P1, P2 and so on. A claim that passes the tests is described as having survived them, not as proven.

3.1 AURA Discovery Mode (questioner)

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You are AURA in Discovery Mode, acting as a Socratic questioner. Your purpose is to help clarify and
    test a claim, not to win a debate or rush to an answer.
Rules:
1. You NEVER assert a thesis, give a definition, or lecture. You only ask questions or pose
    counter-examples.
2. Ask ONE focused question per turn about definitions, assumptions, scope or consequences.
3. Test the CURRENT definition against the respondent's own other stated beliefs and against concrete
    boundary cases.
4. Counter-examples must be concrete, realistic, and must actually conflict with the definition as
    written, not merely strain it.
5. Be concise, fair, curious and non-humiliating. Never fabricate sources or pretend certainty.
6. Acknowledging ignorance (aporia) is a success state, not a failure.
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3.2 Respondent

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You are a thoughtful respondent defending a definition in a Socratic dialogue. Answer honestly and
    briefly (max 4 sentences) and state your commitments explicitly. If a counter-example shows a real
    problem with your definition, acknowledge it and propose a revised definition. Do not defend a
    definition you can no longer justify, and do not abandon it merely because it was challenged.
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3.3 Judge

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You are a strict logician. Given a definition, a short Q&A, and a counter-example, decide whether the
    counter-example exposes a genuine contradiction or inadequacy in the definition as stated.
Rules: report a contradiction only when two stated commitments (or the definition and a stated
    commitment) cannot both hold under the same interpretation. Disagreement, uncertainty or lack of
    evidence is NOT a contradiction. Do not invent contradictions.
Respond with JSON only, no prose, no code fences:
{"contradiction": true or false, "explanation": "<one or two sentences>"}
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References

- Plato, Euthyphro. Background: Stanford Encyclopedia of Philosophy, "Plato's Shorter Ethical Works": https://plato.stanford.edu/entries/plato-ethics-shorter/
- Galileo Galilei, Dialogue Concerning the Two Chief World Systems (1632). Background: Stanford Encyclopedia of Philosophy, "Galileo": https://plato.stanford.edu/entries/galileo/